

LAB 5 Gate Sizing

Technology: 22nm PTM-HP

Nominal VDD = 0.8 V

$k = 1.3$ (W_p/W_n), $W_n = 44\text{nm}$, $W_p = k \cdot W_n$ (from Lab 1)

Identify how many inverters and what sizes to drive a large capacitance like 1pF or 10pF.

Answer is a chain of inverters, each inverter α times larger than last inverter.

Lab 5 Step A

Modeling load capacitance as an inverter

$M = C/C_{\min}$

$C_{\min}$ = capacitance of min sized inverter

$C_{\min} = 71.337 \text{ aF}$

C (pF)	M
1	14018
10	140180

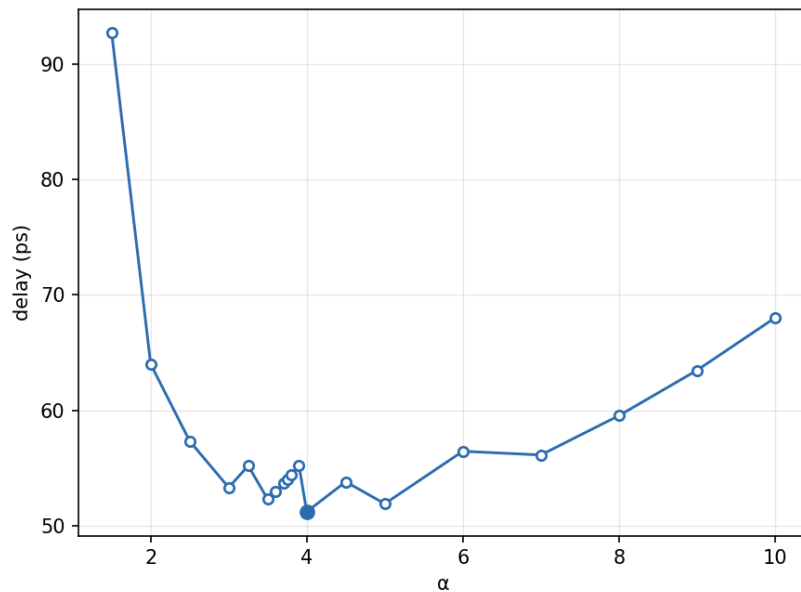
A load of 1pF is equivalent to an inverter of size ~ 14000 the min sized inverter.

Lab 5 Step B

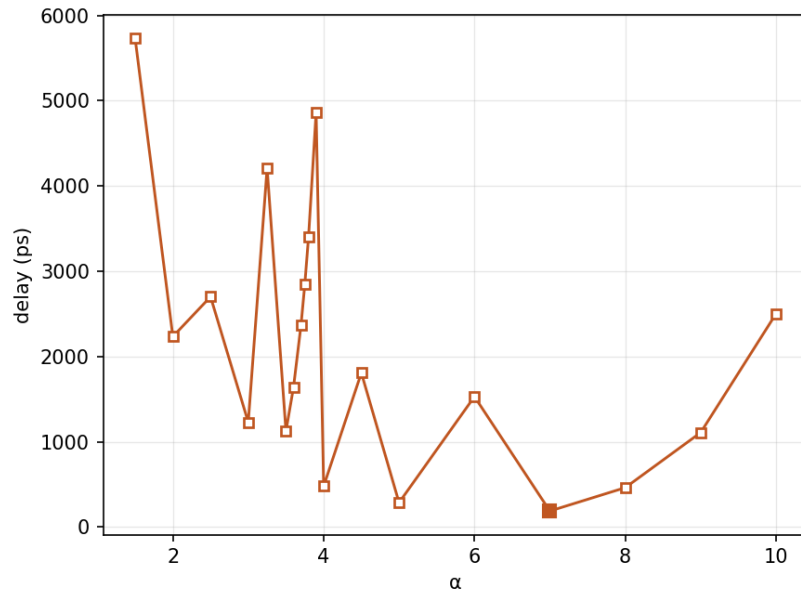
Plot of delay against α

Delay is propagation delay from input of first inverter to input of last inverter (Vdd/2 fall on first inverter to Vdd/2 rise/fall on input of last inverter)

Load = 1 pF Modifying M

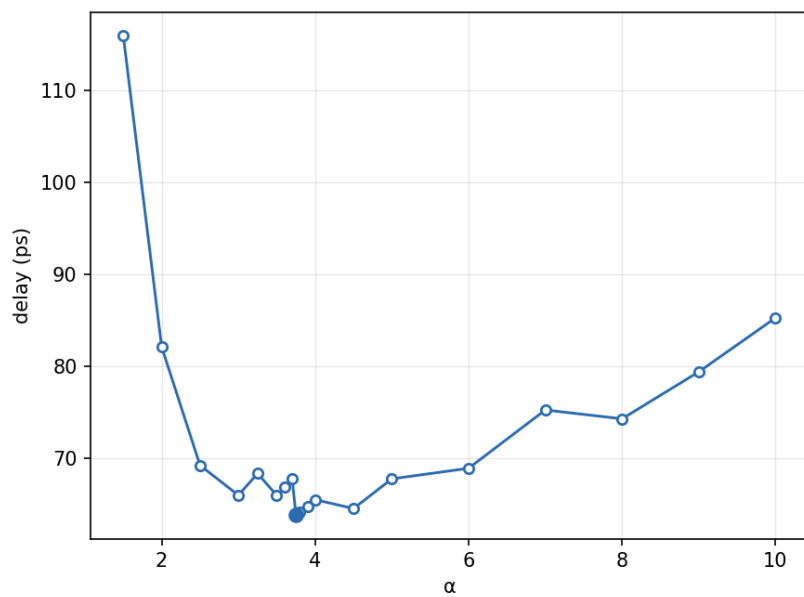


Modifying W

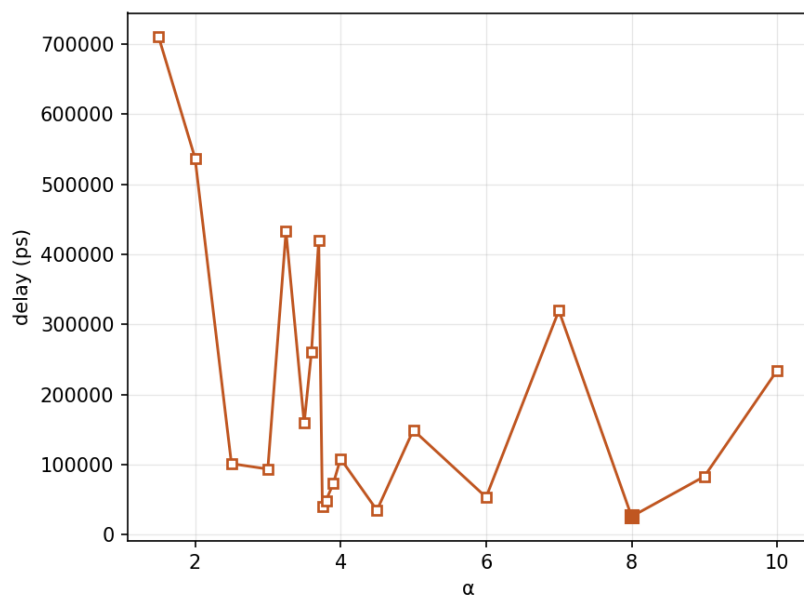


α	N	tpd, sized by M (ps)	tpd, sized by W (ps)
1.5	24	92.67	5729
2	14	64.01	2234
2.5	11	57.35	2701
3	9	53.33	1229
3.25	9	55.22	4208
3.5	8	52.31	1125
3.6	8	52.98	1638
3.7	8	53.69	2371
3.75	8	54.06	2845
3.8	8	54.44	3408
3.9	8	55.26	4861
4	7	51.24	478.9
4.5	7	53.83	1808
5	6	51.93	286.8
6	6	56.47	1530
7	5	56.15	188.2
8	5	59.55	462.4
9	5	63.47	1108
10	5	68.01	2502
14000	1	15780	10970

Load = 10 pF Modifying M



Modifying W



α	N	tpd, sized by M (ps)	tpd, sized by W (ps)
1.5	30	115.9	710500
2	18	82.07	537500
2.5	13	69.19	101500
3	11	65.94	93700
3.25	11	68.3	432800
3.5	10	65.93	159300
3.6	10	66.79	261000
3.7	10	67.76	420200
3.75	9	63.82	39670
3.8	9	64.1	48760
3.9	9	64.7	73180
4	9	65.46	108500
4.5	8	64.49	34950
5	8	67.72	149300
6	7	68.87	53540
7	7	75.21	319900
8	6	74.25	25940
9	6	79.36	83220
10	6	85.23	234600
140000	1	155300	2.381